

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	XXXX
Project Title	AI Debt Settlement System
Maximum Marks	5 Marks

Project Proposal

The proposed **AI Debt Settlement System** aims to help individuals effectively manage their loans and financial obligations using Artificial Intelligence. The system provides personalized debt analysis, financial health assessment, and AI-powered settlement recommendations to assist borrowers in making informed financial decisions. It addresses the challenges of manual debt management, improves financial planning, and reduces the burden of loan repayment through intelligent insights.

Project Overview

Objective

The primary objective of this project is to develop an AI-powered debt settlement platform that enables users to manage multiple loans, monitor financial health, assess debt stress, and receive personalized settlement recommendations and negotiation strategies based on their financial profile.

Scope

The project provides a complete web-based debt management solution that includes:

- User Registration and Secure Login
- Loan Management (Add, Edit, Delete, Search Loans)
- Financial Dashboard with Key Metrics
- AI-Based Debt Stress Analysis
- Personalized Settlement Recommendations
- Negotiation Strategy Generation using Google Gemini AI
- User Profile Management
- PDF Report Generation
- Secure JWT Authentication

Problem Statement

Description

Many borrowers struggle to manage multiple loans due to increasing EMIs, overdue payments, and limited financial knowledge. Existing systems primarily record loan information but do not provide intelligent financial guidance or personalized debt settlement recommendations. This often leads to poor financial decisions, increased debt burden, and financial stress.

Impact

Solving these issues will:

- Improve personal financial planning.
- Reduce debt-related stress.
- Help borrowers understand their financial health.
- Enable informed debt settlement decisions.
- Reduce loan default risk.
- Promote responsible financial management.
- Improve overall user confidence and financial stability.

Proposed Solution

Approach

The proposed system combines modern web technologies with Artificial Intelligence to analyze users' loan details and financial information. It securely stores user data, calculates important financial metrics, evaluates debt stress, and uses Google Gemini AI to generate personalized settlement recommendations and negotiation strategies.

The platform also provides a user-friendly dashboard, loan management system, and downloadable reports for better financial planning.

Key Features

- Secure User Registration and Login using JWT Authentication
- Loan Management (Add, Update, Delete, Search)
- Interactive Financial Dashboard
- Debt Stress Analysis
- Financial Health Score Calculation
- AI-Powered Settlement Recommendations
- AI-Based Negotiation Strategy Generation
- User Profile with Personalized Statistics

- PDF Settlement Report Generation
- Responsive User Interface using React and Tailwind CSS
- REST API developed with FastAPI
- Secure Database Management using SQLite and SQLAlchemy

Expected Benefits

- Faster debt analysis
- Personalized financial guidance
- Reduced financial stress
- Improved repayment planning
- Better decision-making using AI insights
- Secure management of user financial data
- Enhanced user experience through automation

Technologies Used

- **Frontend:** React.js, Tailwind CSS, Axios
- **Backend:** FastAPI (Python)
- **Database:** SQLite
- **ORM:** SQLAlchemy
- **Authentication:** JWT, Passlib (bcrypt)
- **AI Service:** Google Gemini AI API
- **Report Generation:** jsPDF
- **Version Control:** Git & GitHub

